A · Clinical need versus image physics

1 Regional function is a distinct measurement problem

Ejection fraction compresses 3-D deformation into a chamber ratio; compensation, cancellation and timing differences are hidden.

§1

2 Cine carries no material-point identity

Tagging imprints a pattern and DENSE encodes displacement in phase; bSSFP myocardium is homogeneous, so identity is not in the signal.

§2 · Fig. 2

↓ *Leap 1: physics → estimation*

B · From physical gap to dependent number

3 Correspondence is inferred, and the prior selects

Many deformation fields fit the same images; smoothness, incompressibility, biomechanics or a training distribution picks one.

§2–3 · Fig. 1a · Table 1

4 A reported value depends on three choices

Which estimator inferred it, how strain was defined, what it was validated against: ‘is cine FT regional strain accurate’ is incomplete.

§2

C · Evidence audit: what each endpoint supports

5 Boundary endpoints cannot confirm dense correspondence

Dice, surface distance, field regularity and sparse landmarks measure boundary geometry and sampled points; identical borders admit different interior motion.

§4 · Table 2

↓ *Leap 2: reported endpoint → supported claim*

6 Each validation target supports one claim and leaves one gap

Known motion, paired DENSE or tagging, scan–rescan, LGE, outcome; software, sampling, strain convention, statistical unit and reference uncertainty are stated before attribution.

§4–5 · Fig. 1b · Fig. 3

7 Paired studies support a threefold qualification

Circumferential strain, global or whole-slice mid-ventricular scale, matched definitions; agreement varies with component, scale and software, radial least.

§4 · Tables 2–3

8 Focal attributes are rarely measured directly

Magnitude, location, extent and timing; the MRXCAT2.0 test reports whole-slice errors without scar-level recovery; equal means can hide a focal deficit.

Fig. 4 · Table 3 · Fig. 5c

D · Engineering the claim

9 A measurement contract makes the claim falsifiable

Acquisition, estimator, population, spatial unit, strain definition and reference state, time point, comparator, prespecified tolerance.

§6

10 No single agreement index validates all of it

Known motion: capacity and technical error. Paired material reference: material-sensitive agreement. Scan–rescan: precision. Tissue and decisions: relevance and impact.

§6 · Fig. 5 · Table 4